Answers and talk notes

Math is Beauty

What your student is learning

Tonight we use bilateral symmetry — two sides that are mirror images — to describe shapes in nature and in things people build.

Answers

Ask your student to explain before you look.

1. The two sides of the butterfly match, so the butterfly is symmetric.
2. The left wing of the butterfly is a mirror image of the right wing.
3. A butterfly has bilateral symmetry. Its two sides mirror each other.
4. A line down the middle of the butterfly's body is its line of symmetry.
5. The matching wings give the butterfly balance. Neither side looks heavier.

Talk about it

If you fold on your line, do the edges land exactly on each other?

Home mission

Fold a piece of paper in half. Cut half of a butterfly shape along the fold, then open it. Ask your student: where is the line of symmetry? Then cut a lopsided shape and fold it to test.

When they are stuck, ask

- Where could you fold it so both halves match?
- Does the left side look like the right side?
- Is there more than one line that works?

Try not to say

- Everything has symmetry somewhere.
- Close enough counts.
- Symmetry is only about left and right.